CHAPTER 10

VECTOR ALGEBRA

CHAPTER ESSENCE AT A GLANCE:
Vector algebra, dot product (work, projection), cross product (area, torque), scalar triple product, and unit vector resolution.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Vector Magnitudes and Direction Cosines:** Vectors possess magnitude and orientation in space; dividing a non-zero vector by its magnitude yields a unit vector, whose components along coordinate axes represent direction cosines whose sum of squares equals one.

• **Scalar Product and Perpendicularity:** The dot product of two vectors yields a scalar quantity equal to the product of their magnitudes and the cosine of their included angle, vanishing if and only if two non-zero vectors are mutually perpendicular.

• **Vector Product and Orthogonality:** The cross product produces a vector perpendicular to both operand vectors with magnitude proportional to the sine of their included angle, vanishing when non-zero vectors are collinear.

• **Vector Projection and Area Applications:** The scalar projection of vector a onto vector b equals their scalar dot product divided by the magnitude of b, while the magnitude of the cross product computes the area of the spanned parallelogram.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}, \quad l^2 + m^2 + n^2 = \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

Unit vector normalization and sum of squares of direction cosines

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta \rightarrow \cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} \quad (\mathbf{a} \perp \mathbf{b} \Leftrightarrow \mathbf{a} \cdot \mathbf{b} = 0)$$

Scalar dot product and mutual perpendicularity condition

$$\text{proj}_{\mathbf{b}}\mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|}$$

Scalar projection of vector a onto vector b

$$\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2)\hat{\mathbf{i}} - (a_1b_3 - a_3b_1)\hat{\mathbf{j}} + (a_1b_2 - a_2b_1)\hat{\mathbf{k}}$$

Vector cross product determinant and included sine angle

$$A_{\text{triangle}} = \frac{1}{2}|\mathbf{a} \times \mathbf{b}|, \quad A_{\text{parallelogram}} = |\mathbf{a} \times \mathbf{b}|$$

Planar areas spanned by adjacent vectors a and b

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** [Trap Alert: Projection Denominator] Projection of a on b has |b| in the denominator, NOT |a|! Projection is a scalar quantity.

• **Alert:** [Trap Alert: Cross Product Order] Cross product is anti-commutative: a × b = - (b × a). Direction is given by right-hand thumb rule.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define: (a) Unit vector, (b) Collinear vectors, (c) Equal vectors with examples. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Definitions]** 1. Unit Vector: A vector whose magnitude is exactly unity (1): hat(a) = a over |a|. 2. Collinear Vectors: Two or more vectors that are parallel to the same straight line, irrespective of their magnitudes and directions (b = λ a). 3. Equal Vectors: Two vectors having identical magnitudes and the same direction, regardless of initial positions. [2 Marks]

• **[Boxed Result: Unit (|a|=1), Collinear (parallel: b=λa), Equal (same magnitude & direction)]**

2. Find direction cosines and direction ratios of vector r = i - 2j + 3k. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Direction Ratios & Direction Cosines]** r = hat(i) - 2 hat(j) + 3 hat(k). Direction ratios: a = 1, b = -2, c = 3. [1 Mark]

• Magnitude: |r| = sqrt(1^2 + $(-2)^2$ + 3^2) = sqrt(1 + 4 + 9) = sqrt(14). Direction cosines: l = a over |r| = 1 over sqrt(14). m = b over |r| = - 2 over sqrt(14). n = c over |r| = 3 over sqrt(14). [1 Mark]

• **[Boxed Result: Direction ratios: 1, -2, 3; Direction cosines: 1 over √14, -2 over √14, 3 over √14]**

3. If |a + b| = |a - b|, prove that vectors a and b are perpendicular. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Proof]** Given |a + b| = |a - b|. Squaring both sides: $|a + b|^2$ = $|a - b|^2$. [1 Mark]

• (a + b) · (a + b) = (a - b) · (a - b). $|a|^2$ + 2 (a · b) + $|b|^2$ = $|a|^2$ - 2 (a · b) + $|b|^2$. Canceling $|a|^2$ + $|b|^2$: 2 (a · b) = - 2 (a · b) implies 4 (a · b) = 0 implies a · b = 0. Since a · b = 0, vectors a and b are PERPENDICULAR. Proved. [2 Marks]

• **[Boxed Result: a · b = 0 ⇒ a ⊥ b]**

4. Find unit vector perpendicular to both vectors a = i - j + k and b = 2i + 3j - k. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Cross Product Calculation]** a = hat(i) - hat(j) + hat(k), b = 2 hat(i) + 3 hat(j) - hat(k). a × b = |[[hat(i), hat(j), hat(k)], [1, -1, 1], [2, 3, -1]]| = hat(i)(1 - 3) - hat(j)(-1 - 2) + hat(k)(3 - (-2)) = - 2 hat(i) + 3 hat(j) + 5 hat(k). [1.5 Marks]

• **[Unit Vector Formulation]** |a × b| = √($(-2)^2$ + 3^2 + 5^2) = √(4 + 9 + 25) = √38. Unit vector hat(n) = ± (a × b) over |a × b| = ± (- 2 hat(i) + 3 hat(j) + 5 hat(k)) over √38. [1.5 Marks]

• **[Boxed Result: hat(n) = ± (- 2 hat(i) + 3 hat(j) + 5 hat(k)) over √38]**

5. Find the area of triangle with vertices A(1, 1, 2), B(2, 3, 5), and C(1, 5, 5) using vectors. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Side Vectors]** A(1, 1, 2), B(2, 3, 5), C(1, 5, 5). AB = (2 - 1) hat(i) + (3 - 1) hat(j) + (5 - 2) hat(k) = hat(i) + 2 hat(j) + 3 hat(k). AC = (1 - 1) hat(i) + (5 - 1) hat(j) + (5 - 2) hat(k) = 0 hat(i) + 4 hat(j) + 3 hat(k). [1 Mark]

• **[Cross Product & Area]** AB × AC = |[[hat(i), hat(j), hat(k)], [1, 2, 3], [0, 4, 3]]| = hat(i)(6 - 12) - hat(j)(3 - 0) + hat(k)(4 - 0) = - 6 hat(i) - 3 hat(j) + 4 hat(k). |AB × AC| = √($(-6)^2$ + $(-3)^2$ + 4^2) = √(36 + 9 + 16) = √61. Area of Triangle = (1 over 2) |AB × AC| = √61 over 2 sq. units. [2 Marks]

• **[Boxed Result: Area of Triangle = √61 over 2 sq. units]**

6. Find the area of parallelogram whose adjacent sides are a = 3i + j + 4k and b = i - j + k. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Cross Product of Adjacent Sides]** a = 3 hat(i) + hat(j) + 4 hat(k), b = hat(i) - hat(j) + hat(k). a × b = |[[hat(i), hat(j), hat(k)], [3, 1, 4], [1, -1, 1]]| = hat(i)(1 - (-4)) - hat(j)(3 - 4) + hat(k)(-3 - 1) = 5 hat(i) + hat(j) - 4 hat(k). [1.5 Marks]

• **[Area of Parallelogram]** Area = |a × b| = √(5^2 + 1^2 + $(-4)^2$) = √(25 + 1 + 16) = √42 sq. units. [1.5 Marks]

• **[Boxed Result: Area of Parallelogram = √42 sq. units]**

7. If a, b, c are unit vectors such that a + b + c = 0, find a·b + b·c + c·a. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup & Expansion]** Given a + b + c = 0, and |a| = |b| = |c| = 1. $|a + b + c|^2 = 0$. [1 Mark]

• $|a|^2 + |b|^2 + |c|^2 + 2 (a \cdot b + b \cdot c + c \cdot a) = 0$. $1^2 + 1^2 + 1^2 + 2 (a \cdot b + b \cdot c + c \cdot a) = 0$. 3 + 2 (a · b + b · c + c · a) = 0. a · b + b · c + c · a = - 3 over 2. [2 Marks]

• **[Boxed Result: a · b + b · c + c · a = - 3 over 2]**